

At a glance

Curtain walling may be defined as a weatherproof, non-load-bearing enclosure of windows or metal panels in a light metal framework that is fixed back to the structure of a building, usually at intermediate floor levels. There are various systems in common use ranging from simple, site-assembled 'stick systems' to larger, pre-assembled panels or unitised systems. See [Introduction](#) .

Many glass curtain walls comprise a system of mullions and transoms with the insulating glass units held in place with clamping plates. In more recent years, silicone bonded systems have become popular; in these systems either 2 or 4 edges of the glass are held in place with silicone adhesive. See [Methods of preventing water ingress](#) .



Few, if any, curtain walling systems can be made totally weathertight. It is usual to accept that some leakage will take place from the outer, weather facing seals, but to drain this water away in a controlled fashion. There are several mechanisms of water penetration into a system: force of gravity, kinetic energy, capillarity, pressure assisted capillarity, surface tension and air pressure differentials. Of these, air pressure differentials are possibly the most significant. See [Weathertightness](#) .

Wind pressure against a facade creates localised areas of high pressure. If the air pressure inside a building is lower, air leaks through, taking moisture with it. If the air in the cladding system can equalise quickly with the external air pressure, the propensity for leakage is much reduced. Drained and pressure equalised cladding systems (which are generally most common) use this principle, and divide the ventilated void in the cladding into small fields to permit rapid equalisation. See [Problems with pressure plates](#) .

Problems of leakage are possibly the most common form of defect, although defective fixing details are not unknown. Fixings must be robust and durable, as they are inaccessible and cannot be inspected or repaired easily. The fixing system must be capable of accommodating building movements due to loading, creep, thermal and moisture movements, etc. See [Failure modes](#) .

Most systems rely to a large extent upon the performance of sealants. The basic types of sealant failure are:

- adhesion failure;
- cohesion failure;
- delamination failure; and
- reversion.

See [Identifying defects](#) .

Introduction

In historical terms a curtain wall formed the enclosure to a fortification.

The current use of the term describes a weatherproof, non-load-bearing enclosure of windows or metal panels in a light metal framework that is fixed back to the structure of a building, usually at intermediate floor levels. Curtain walling is a form of cladding, providing a weatherproof enclosure and increasingly serving as a means of controlling the internal environment of a building.

The science and technology of cladding is fairly complex. Therefore the cladding industry is often plagued with design or workmanship problems. Over recent years, dozens of cladding manufacturers and installers have failed, often as a result of difficulties with particular projects. One major cladding manufacturer is reputed to have purchased an entire office building from the employer as a means of resolving a dispute. It is unsurprising therefore that many surveyors shy away from cladding inspections, preferring to leave the inspection and resolution of defects to those with appropriate skills.

The cladding supply chain is also complicated - populated with a large number of specialist contractors of varying degrees of competence and a smaller number of correspondingly larger manufacturing organisations. Frequently (and especially with window systems) the components of the system are supplied by the manufacturer to a franchised fabricator, who may decide to introduce non-franchised components into the system and produce an installed product that is not fully in accordance with the manufacturer's specification or design.

There may be an infinite number of variations of cladding designs, but there are some fairly basic concepts that are worth examining as they help develop an understanding of how cladding works and why it can go wrong.

Types of curtain walling system

Stick system

Stick construction is the traditional form of curtain walling, comprising a grid of mullions and transoms into which various types of glass and/or insulated panels can be fitted. The systems are usually fairly low cost, using extruded components supplied by the system manufacturer. Mullions are usually assembled first, with transoms then fitted in-situ.

A variety of jointing systems are employed but most stick systems have some form of stub connection between individual lengths or transoms. Mullions are usually fixed back to the structure at floor levels, although longer spans are possible according to the section properties of the component.

Aluminium is the usual material for the framework, although some older buildings experimented with steel. Most of the grid assembly work is done on site. This permits some measure of dimensional adjustment and tolerance, but it is highly workmanship-sensitive and relies on accurate cutting of components and the correct use of sealants at joints.

Assembly on site is fraught with risk as many systems are superficially similar so that a contractor who is familiar with one system may omit crucial components such as seals at junctions when assembling components provided by another manufacturer. ([Weatherproofing problems in modern glazing systems](#) , CIOB, 1992) Similarly, if shop drawings are hard to read (perhaps in a different language) or do not detail items such as drainage or ventilation slots, there is a risk that these items are neglected or installed incorrectly.

Unitised system

Unitised systems comprise narrow-width, storey-height units of aluminium framework containing glazed and/or opaque insulated panels. The entire system is pre-assembled under factory-controlled conditions, thus avoiding some of the problems of assembly encountered on site.

The unitised panels are usually craned into position, with pre-positioned brackets attached to the floor slab or the structural frame. Modern installation techniques increase the speed of erection and often minimise the requirement for scaffolding.

Unitised systems do have higher direct costs but nowadays the curtain walling to most prestige buildings is of this type.

Panellised systems

Like unitised systems, panellised curtain walling is constructed in the factory to improve quality control and speed of erection. However, the systems comprise large prefabricated panels of bay width and storey height, which are connected back to the primary structural columns or to the floor slabs. Panels may be of precast concrete or comprise a structural steel framework and are often clad with a variety of stone, metal and masonry cladding materials.

The advantage of these systems is improved adherence to specification as a consequence of factory prefabrication, allowing improved control of quality and rapid installation with the minimum number of site sealed joints. Panellised systems are less common and more expensive than unitised construction.

Structural silicone

Structural sealant glazing is a form of glazing that can be applied to stick or unitised curtain walling systems. With structural sealant glazing the insulating glass units (IGUs) are attached to the grid framework with factory applied structural silicone sealant rather than by pressure plates and gaskets in a more traditional system.

Structural glazing

Structural glazing typically comprises large thick single panes of toughened glass assembled with special bolts and brackets that are supported by a secondary steel structure. This form of glazing is often referred to as 'Planar' glazing and is commonly used to form the enclosure to atriums and entrances. [Planar](#) is a trade name rather than a generic type of glazing system.

Rainscreen systems

Rainscreens are not curtain walls as such, as they provide a screening function rather than an enclosure in its own right. A rainscreen is used to shield a wall, whether this be of masonry, metal studwork or in some cases glass. Usually a rainscreen is designed to permit some controlled leakage with the main functions of resistance to air and water being provided by the

shielded wall behind.

Weathertightness

Patterns of wind and rain exposure

All openings in a facade are a potential source of water penetration. However, the geometry of the building, its height and orientation all affect the performance of the facade in terms of air and water leakage.

Walls facing the wind are subjected to greater driving rain, while leeward walls are more sheltered. However, variations in pressure occur between the sides, centre and top of a wall. As an air current meets a building, it splits to travel round and over it.

This parting of the air creates a cushion of high pressure but relatively still air at the centre of the wall, which tends to reduce the amount of rain hitting it at that location. By contrast, the amount of rain hitting the wall at the corners or top of the building can be as much as 20-50 times that at the centre, since the movement of air accelerates around these features. A tall, narrow building will suffer more from this type of problem than a wider, more squat building.

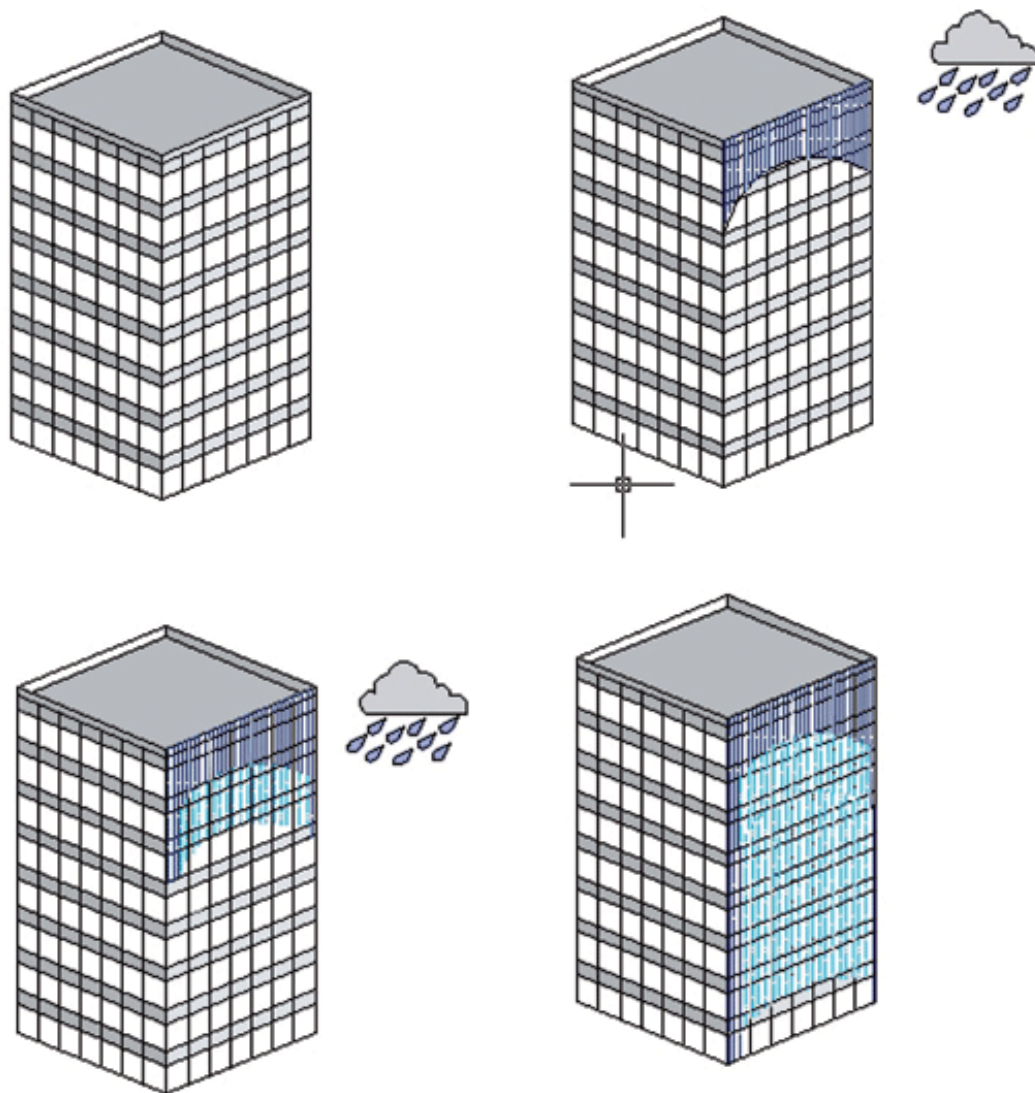


Illustration of rainfall effect on a facade

The diagram above shows the effects of rainfall on the facade of a tall building. After about 10 minutes the migration of wetting commences, gradually extending down the building and down the corners. With impervious claddings, such as metal and glass curtain walls, water simply flows down the wall surface and the accumulated flow can be significant by the time it reaches the bottom of a tall building. Wind flow around corners and parapets can also draw water laterally and even upwards.

Aside from appearance and climate control, the key function of a curtain walling system is to exclude air and water. Indeed, most failures are related to water leakage in one way or another and a proper understanding of the mechanisms of water entry and the main ways of preventing this will assist in the diagnosis of leakage problems.

There are 5 mechanisms of water ingress:

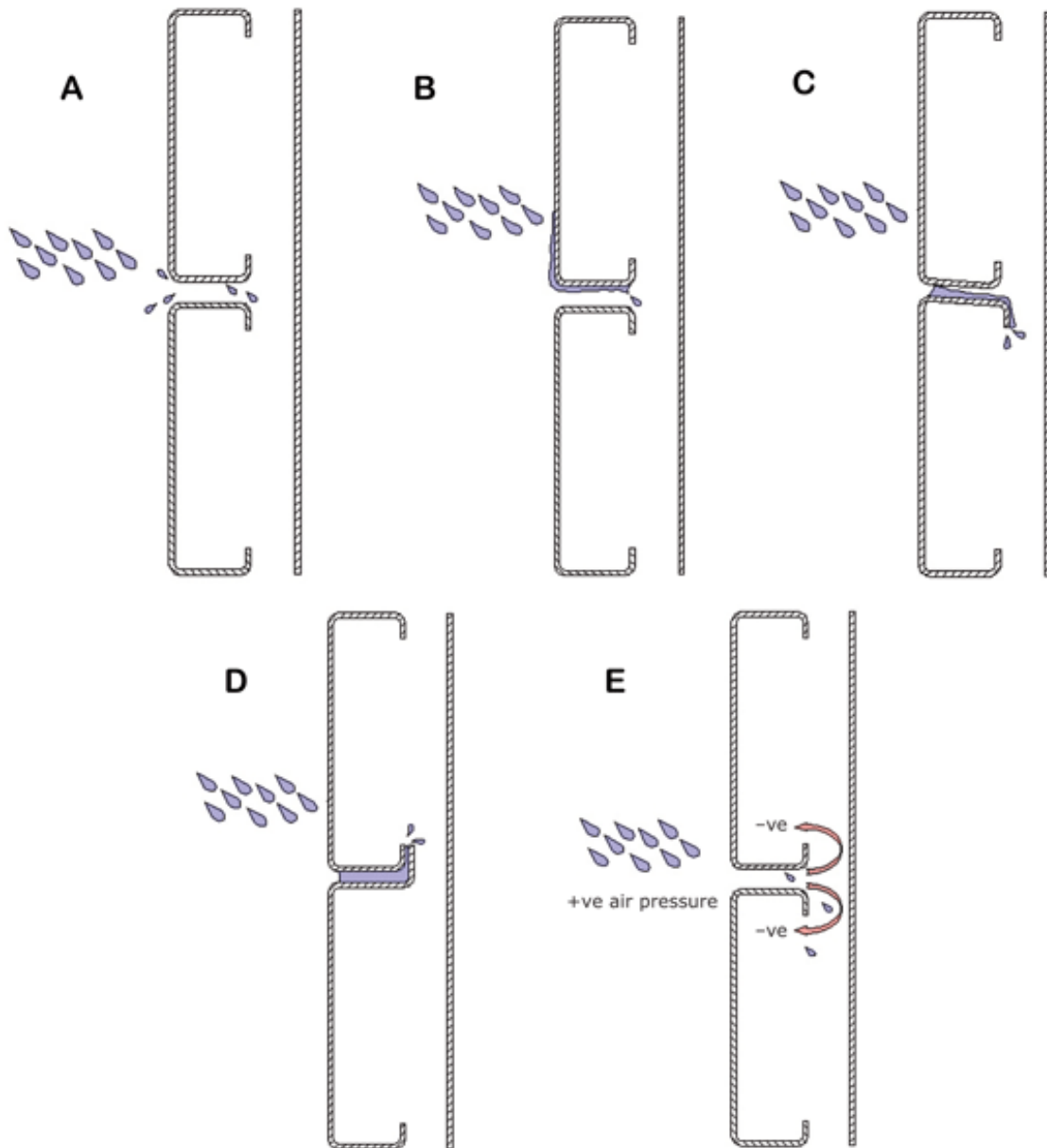
- kinetic energy;
- surface tension;
- capillarity;
- gravity; and
- air pressure differentials.

Essentially, the methods of prevention of water ingress are:

- to use sealants;
- to design joints in such a way as to prevent water leakage;
- to use mechanical seals (gaskets); and
- to use a combination of two or more of the above.

Although it is theoretically possible to construct a curtain wall using only one of the above methods, in practical terms it is usual to employ more than one. Different options are possible:

- fully sealed or front sealed systems (usually where a flat plane is required);
- drained and back ventilated systems; and
- pressure equalised rainscreen systems.



Kinetic energy (figure A)

A rain drop possesses translational kinetic energy; a function of the mass of the raindrop and its speed. When driven by wind, drops of rain may have sufficient forward momentum to be carried through small joints or openings in the cladding system. On hitting a surface the drops can shatter, with the resultant smaller drops carrying further into the cladding system. To counteract this, joints are designed with a baffle so as to provide a labyrinthine structure.

Studies by Choi ([Velocity and impact direction of wind driven rain on building faces](#)) on the effect of wind driven rain showed that drops of small diameter (0.5mm) are capable of moving upwards, particularly in strong wind. Although the upward movement of larger drops (10mm in diameter) is unlikely, the horizontal longitudinal velocity component can be quite large and raindrops can move in a direction that is horizontal and sideways parallel to the building. This means that not only can raindrops penetrate into the depth of a cavity in a cladding system,

they can also travel sideways within it.

Surface tension (figure B)

Surface tension is the property of a liquid to behave as if it were covered with a weak elastic skin and is caused by the cohesive forces of molecules at the surface. Water has a particularly strong surface tension, which enables it to run vertically down a surface and then back horizontally over an overhanging edge. By this mechanism, water can be drawn into a joint that is not exposed to direct rainfall.

To counteract the effects of surface tension, it is usual to provide a small drip on the underside edge of a projection.

Capillarity (figure C)

Like surface tension, capillarity is due to forces at molecular level and an unbalanced attraction between the molecules of the liquid and those of a narrow tube or space. If the molecules of the liquid are more attracted to the tube than other liquid molecules, the liquid will rise. For this to happen, the walls of the tube or enclosure must be sufficiently close together. But capillarity by itself can never result in leakage, simply because water can never rise beyond the faces of the enclosure.

However, wind-assisted capillarity can result in water penetration, as the force of the wind can push water further up the walls of the enclosure and beyond. To counteract capillarity, joints can incorporate an air space to break the surface tension and prevent the water rising at the interfaces.

Capillaries of less than about 0.01mm draw and hold a small volume of water with such high suction that they seldom contribute to rain penetration. A greater volume of water, however, is held by the lower suction in large capillaries such as unbonded interfaces. Large capillaries are important contributors to water penetration when an additional force of even low magnitude is added.

Gravity (figure D)

The force of gravity plays an important role in water penetration as it can pull water down and in through an opening that leads inwards and slightly downwards. While wider joints can prevent capillarity occurring, air within the space coupled with the effects of gravity can result in water travelling vertically until it reaches a horizontal surface.

Air pressure differential (figure E)

Of the 5 mechanisms of water entry described, air pressure differentials are the most important and are often the most significant source of water ingress. Air will move from an area of high pressure to an area of low pressure. This movement of air can carry water with it, so that water can be drawn into a joint, glazing rebate or even to the interior of a building.

Wind blowing against or around a building creates large pressure differentials. The leakage of air through a wall system prevents the outside pressure from equalising across the cladding - rather like trying to inflate a balloon with a hole at both ends, the air leaks away. The key to dealing with air pressure differentials is to aim to equalise pressures as quickly as possible (see Pressure equalised systems under [Methods for preventing water ingress](#)). This is accomplished by providing a rigid air barrier on the inside and suitable gaps on the outside to allow air movement.

Methods of preventing water ingress

Essentially there are 3 options in curtain wall design.

Face sealed systems

Since the predominant materials of curtain walling, steel and glass, are impervious, careful joint design should be able to prevent water ingress. Early curtain walling systems tended to be face sealed and relied on a weatherproof outer seal to prevent water penetration.

The seal must remain completely free of defects to prevent leakage paths occurring. This means that workmanship must be of a high standard. Achieving watertightness during construction is one thing; maintaining it throughout the life of the cladding is something else, when the individual components are subject to temperature and structural movements, ageing of sealants, etc.

In a truly face sealed system there is no provision for drainage: any water that bypasses the outer seal could result in internal water ingress. With single glazing systems it had always been the case that glass would be fully bedded, initially with glazing putty and latterly with various synthetic glazing compounds. Early installations of insulating glass units (IGUs) followed the same route, with the edges of the units fully encapsulated with 'wet' applied glazing compounds.

The systems are now largely obsolete simply because it was very difficult to install the sealant without the creation of voids or air pockets, which, as the building aged could become filled with water. More often than not the voids would be around the edge of the IGU, thus holding moisture in contact with the edge seal and leading to premature breakdown of the unit allowing it to mist up. Furthermore, wind-induced deflections or movements in the glass could pump the water contained in the voids, forcing it towards the interior of the building.

A variation on the wet sealed system was the zipper gasket design. In this form of installation the IGUs (or single glazed panels) were fitted into neoprene rubber gaskets that completely encapsulated the edge of the glass. The gasket would be designed to hold 2 adjacent panels and contained a groove into which a rigid 'zipper' strip was pushed. The zipper would compress the gasket against the sides of the glass and, theoretically at least, provide a watertight joint.

A small number of proprietary systems employed the zipper gasket principle but they were notorious for leakage problems. Because of their flush, or nearly flush appearance, the design found favour with architects wishing to achieve a flat plane appearance in the design of a facade.

With a zipper system, the glass is restrained by the gasket; there are no additional mechanical supports. Thus, the gasket material must be rigid enough to carry the weight of the glass as well as wind loads without deformation. In practice it is easier to accomplish the structural function than the waterproofing function:

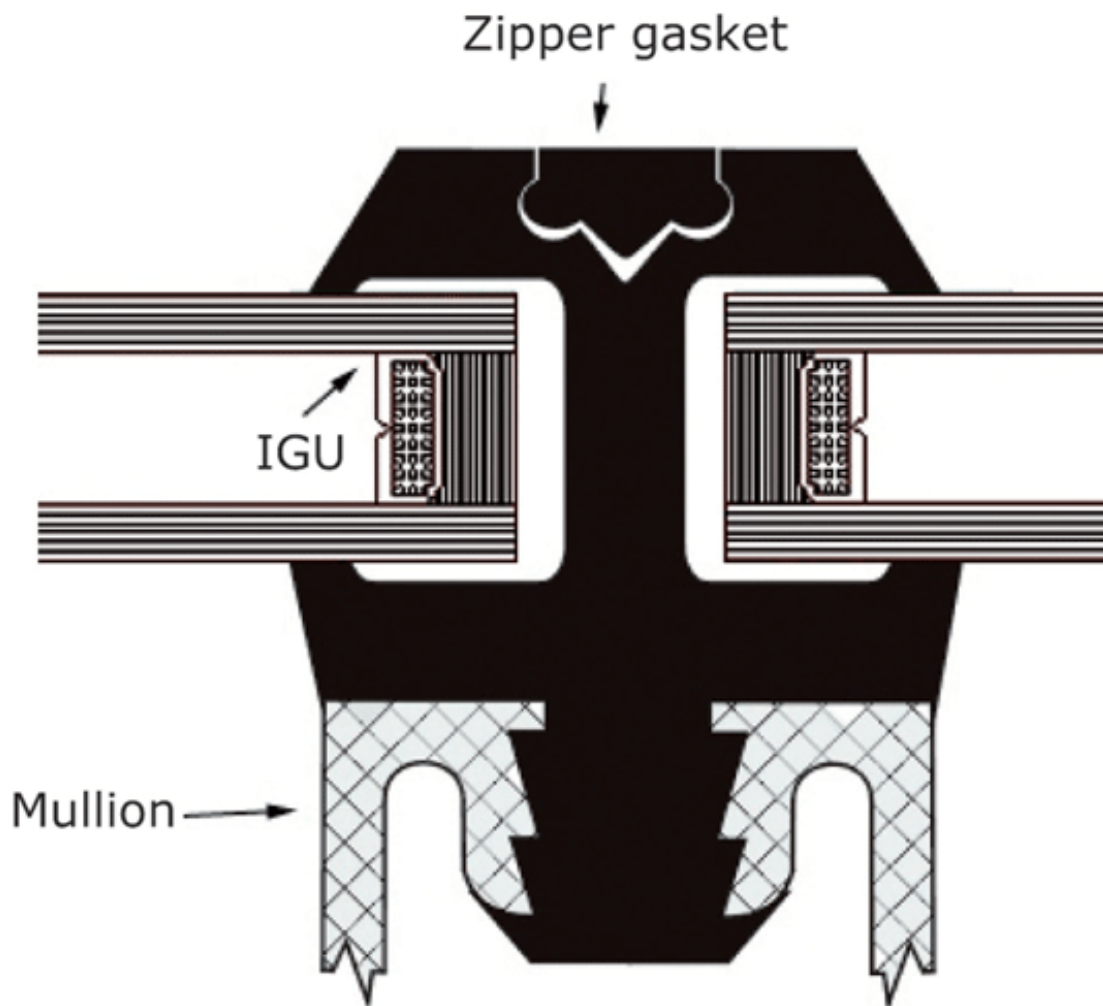
- primarily because it is impossible to install gaskets without joints; and
- secondly because of the difficulty of maintaining adequate compression on the gasket throughout its contact face and throughout its life.

To be effective at excluding water, the gasket must be held at pressure in contact with the glass or panel. The pressure of the gasket at its contact face is called lip pressure. Over time, lip pressure decreases as a result of normal hardening or weathering, compression set and or shrinkage. When installed, the gaskets are stretched; they tend to revert over time and also shrink as a result of the gassing off of volatile plasticisers (Nicastro, D., *Zipper gaskets*, article in *The Construction Specifier*, The Construction Specifications Institute, Sept. 1996). The designer of the zipper gasket system is faced with a compromise; the gasket material must be rigid enough to fulfil its structural role, yet remain flexible enough to maintain adequate lip pressure.

The zipper (sometimes termed a 'lock-strip') is usually around 10 points harder in durometer (the 'shore A' hardness reading - a measure of the resistance of plastics toward indentation) than the gasket itself and exerts a force on the centre of the gasket, which is designed to be transferred to the tips of the gasket in contact with the glass. At a corner or cruciform joint the geometry of the joint reduces the effectiveness of the zipper and hence lip pressure can be compromised. Similar problems occur at butt joints.

One North American manufacturer claims to solve the joint problem by forming factory made units to fit specified openings. Corner pieces and tee joints are made by injection moulding; this injection moulding process forces uncured rubber compound into a mould that joins the gasket ends and simultaneously vulcanises the rubber splice.

A face sealed, single stage design has no back up if water bypasses the primary seal. Air pressure differentials ensure that any small leakage paths are exploited. Confronting the weather head on requires a leap of faith - faith in the engineering ability of the designer, and in the skills and knowledge of the installer. Far better to accept the fact that 100% watertightness cannot be achieved and to make allowances for small amounts of water to both leak in and drain out again.



Typical zipper gasket system

Drained systems

An improvement on the face sealed system is the drained system which contains an air cavity behind the primary outer face. While still vulnerable to leakage resulting from air pressure differentials, the cavity can at least be drained. For this arrangement to work a secondary back seal is needed.

Most cladding designers accept that it is difficult to exclude water and therefore provision for a small amount of leakage can be made within a drained system. Typically these dry glazed systems comprise an outer decorative cover plate, and an aluminium pressure plate with 2 narrow rubber oyster gaskets either side clamped against the glass or insulated infill panel. The pressure plates are screw fixed through a thermal break into the mullion or transom member. A further inner gasket between the glass and the mullion or transom provides the back seal.

In drained and ventilated systems the front gaskets provide an initial barrier. The rebates and cavities are drained and ventilated to the exterior to prevent the accumulation of any water that bypasses the outer seals. Drainage is usually via small holes or slots in the underside of transoms that drain water down through the mullions.

Some systems also incorporate a foil-faced, butyl adhesive tape applied over the transom and mullion nosings directly beneath the pressure plate to serve as a secondary line of defence. While these systems will accommodate a small quantity of water within the glazing rebates, the pressure plate needs to be fixed to the correct torque so that the outer gasket seal forms a good seal against the glass.

Another key issue is the adequacy of the joint design. In stick systems transoms are usually joined to mullions via metal stubs screwed to the mullion and sliding into the hollow transom profile. Since the glazing rebate functions as a drainage path, it is important to ensure that water cannot work its way back to the interior of the cladding system at the mullion/transom joint. Water at this point could track back as a result of pressure differentials and wind assisted capillarity. For this reason, manufacturers often specify a layer of sealant at the joint.

Another variation of joint design involves the use of rubber 'infiltration blocks', which are usually placed at cruciform joints. Being single mouldings, they can help keep water away from the corner joints, but again they invariably need to be sealed properly. A failure to seal properly often results in water leakage internally.

Water that enters the system can be drained either from the transoms via slots, or into the mullion, to be discharged at strategic points, usually with a plastic drainage shoe insert. In those systems that rely on an additional butyl foil layer beneath the outer gasket, check that the tape is cut around the drainage shoe, otherwise water can back up in the system and penetrate the building via a convenient joint.

Back ventilation is a concept mainly applied to rainscreen curtain walling systems. To deal with kinetic energy, joints are usually unsealed, but designed as a labyrinth so that drops of rain shattering on impact will not travel further into the joint. Even so, it is accepted that at times water can penetrate as a result of air pressure differentials, surface tension and gravity, so a drained and ventilated void is provided behind the outer skin. Behind the void may be located a 'technical wall' - providing insulation and a vapour barrier. To provide airtightness, joints in the technical wall would be sealed. The basic function of the rainscreen principle is to deal with all of the mechanisms of water entry.

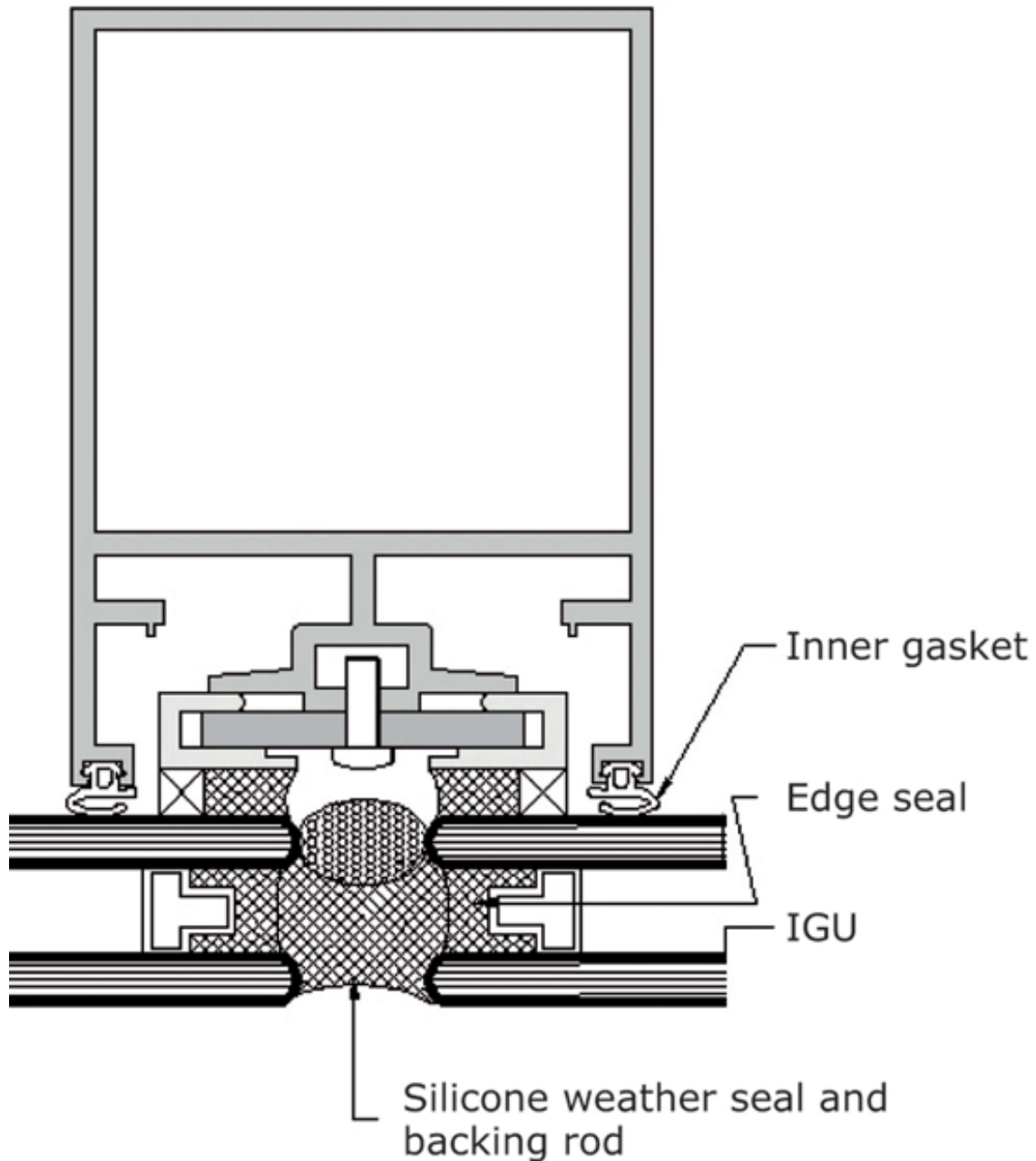
Drained and back ventilated systems are not simply used for rainscreen cladding. Many high-profile office buildings now employ a similar concept - an outer single layer of glass as the rainscreen, a void of perhaps 500-600mm and then an inner glazing system of IGUs. With these systems, positive ventilation is achieved by means of fans and exhaust grills and the air space used either to warm or cool the interior of the building according to season. Although expensive, this method has the advantage that the weather is kept away from the IGUs and so the risk of leakage into the interior of the building is much reduced.

Despite the reservations expressed earlier, one face sealing system that appears to work is the use of structural silicone jointing - a system with a track record of more than 40 years.

Silicone glazing systems (sometimes called SSG) enable a flush elevational treatment with no exposed metal components such as pressure plates or caps. The system also has several

advantages over conventional stick systems, for example:

- it provides a good thermal break;
- it reduces the amount of dirt and water retention in the glazing system;
- it reduces thermally induced stress; and
- it is easy to clean.



An example of a proprietary silicone glazed system in which the IGU is usually factory bonded to a carrier frame with silicone adhesive, in turn fixed to the mullions and transoms with aluminium fixing lugs. In this case, any water that bypasses the outer weather seal between the units can drain away via the air chamber behind.

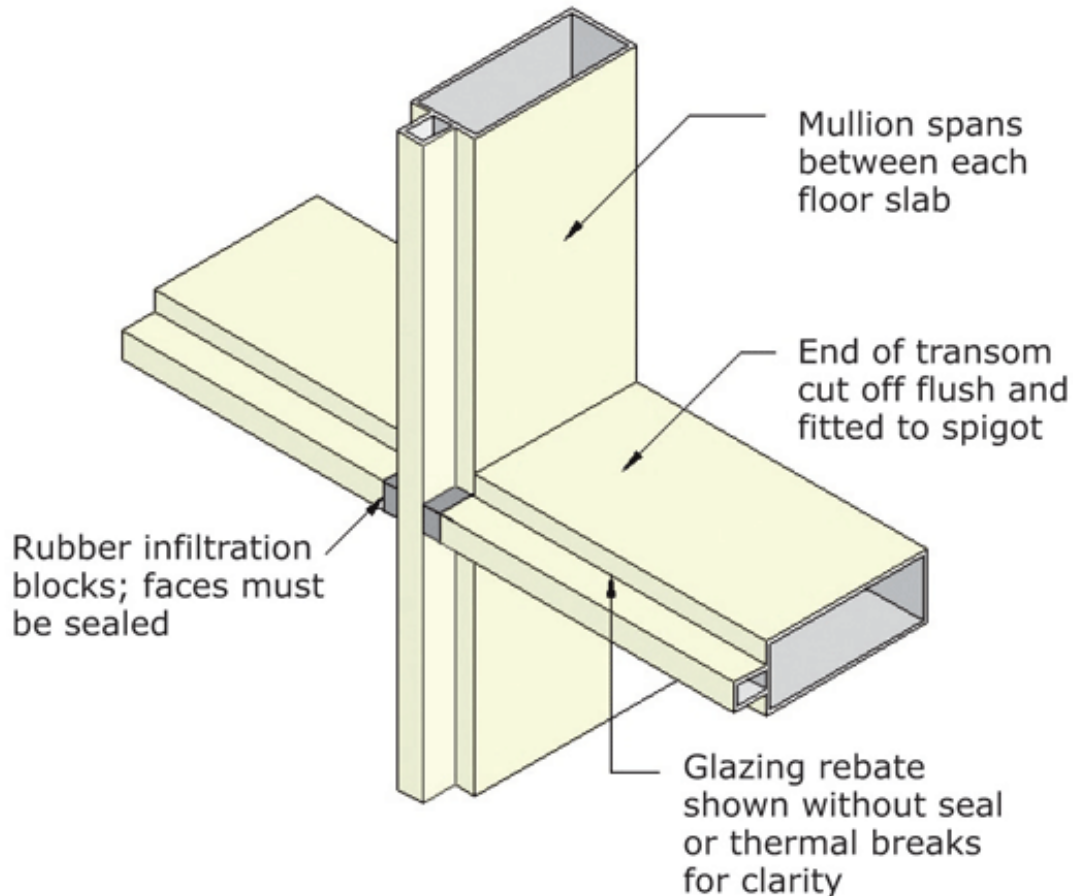
There are 2 basic types of silicone glazing:

- 2-sided; and
- 4-sided.

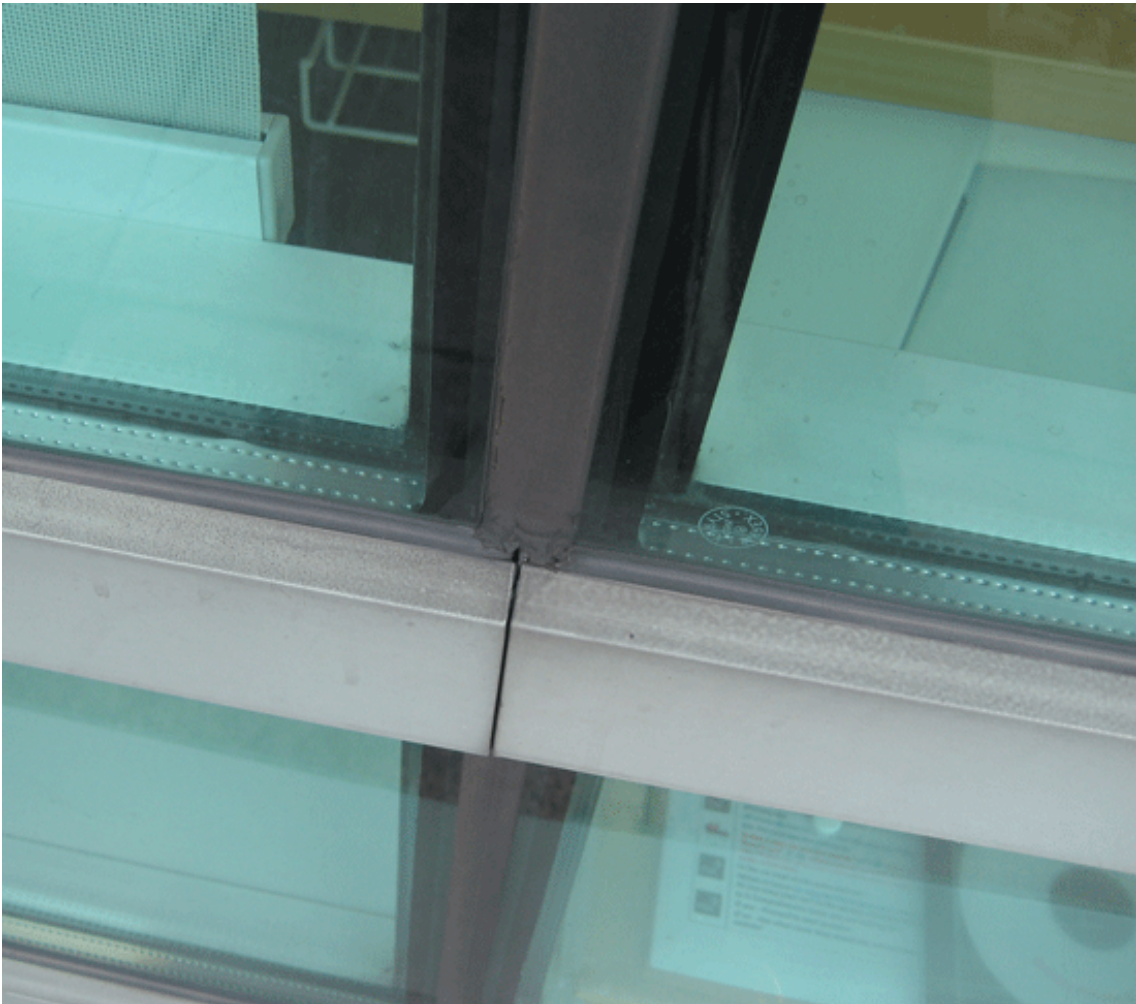
Two-sided glazing provides support to the vertical edges only, while gravitational loads are supported by mechanical means - either a bolted system or a conventional fin support. Two-sided systems should not be confused with butt-joint glazing which does not provide a structural bond to the framing system. Butt-joint glazing provides a weather seal only on 2 edges of the glass.

In 4-sided construction, the glass is bonded on all 4 sides, with the dead weight of the unit being carried by a structural fin and wind loads carried by the silicone joints. Sealant manufacturer Dow Corning suggests that 4-sided glazing systems are less likely to leak due to the adhesive/sealant which blocks air and water ingress. (*Structural Silicone Glazing*, Dow Corning Corporation, USA, 1999)

Not only does the silicone joint carry wind loads it must also accommodate thermally induced movements between the IGU and the structural frame. The weather seal between adjacent units is not designed to be load-bearing. Its function is to exclude air and water, but it must be bonded to both the edges of the glass as well as the edge seal - for this reason the compatibility of sealants must be established, and if necessary demonstrated by test.



Rubber infiltration blocks at junction of mullion and transom



Two-sided silicone glazing - silicone to vertical joints, conventional pressure plate to transoms

Some forms of 2-sided structural silicone glazing are designed to be erected in the field. In this case, the glass is usually held in place with temporary mechanical fixings while curing of the adhesive takes place - this can be as long as 3 weeks. Once the adhesive is cured, the temporary fixings are removed and the weather seal installed. Field installation is fraught with quality control problems and within the UK, there has been a move to factory bonding.

Cleanliness and adequate cleaning with appropriate solvents is essential to avoid loss of adhesion, both in the adhesives and the weather seals. The consequences of adhesion failure of the structural bond could be severe as there are no additional mechanical fixings to hold the glass in place. Catastrophic failure of silicone glazed systems are rare, although worries over structural performance have led to the design of structural spacers - these contain a groove that permits a mechanical clip fixing to the inner leaf.

While the adhesive properties of silicone are well known and documented, problems with compatibility can still occur. In recent years more coated glass products have been marketed in an attempt to improve thermal performance. These coatings are not always compatible with silicone sealants and careful checking is needed to ensure that a suitable bond will be achieved if the coating is to be exposed to the silicone. Similarly, some polyester powder coatings, particularly those with a high gloss, contain a wax residue that may prevent a proper bond.

Adhesion testing is usually by means of a peel test method - although this does not produce actual measurable results. More rigorous pull-off tests can be scientifically controlled and give more reliable results. Pull-off tests usually involve bonding the samples together and then pulling them apart. The sample will fail either in adhesion (one face pulls off) or preferably by cohesion (where the silicone actually remains bonded to the 2 surfaces but splits through its thickness).

Pressure equalised systems

The prime mechanism of water entry into a cladding system is a differential between high and low pressure. If the pressure in the glazing rebate can be equalised with that outside, the movement of air (and hence water) can be eliminated. In practice, there will always be a very slight lag in the equalisation of the pressures in the glazing rebate and the exterior, but the timing can be reduced so that to all intents and purposes the pressures are classed as equalised.

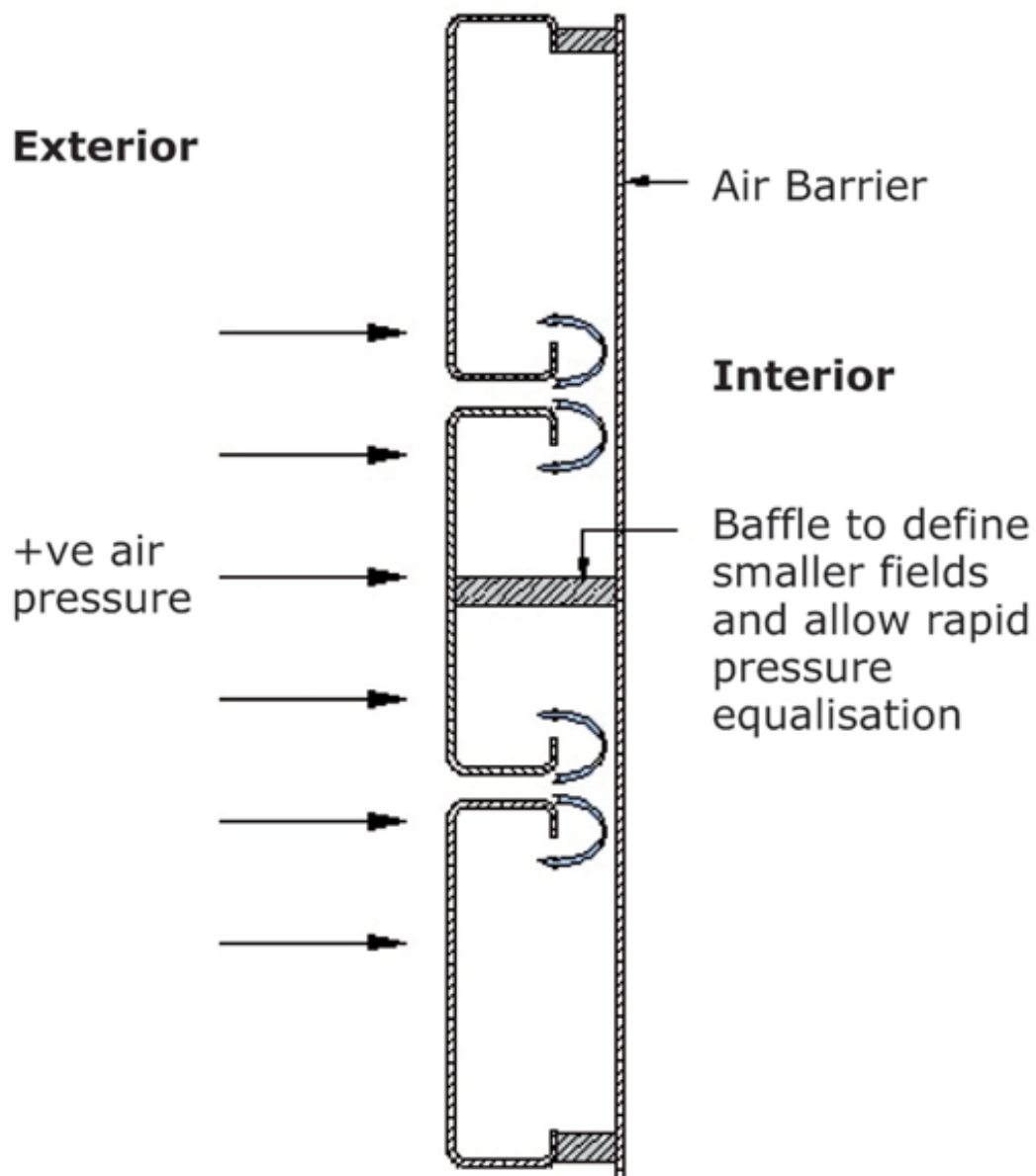
To achieve pressure equalisation it is necessary to:

- provide suitable slots in the pressure plates to enable air pressures to reach equilibrium very quickly;
- divide the facade into zones that are sealed from one another (a zone would normally exist around a single IGU);
- ensure that the rear face of the cladding system is properly sealed to both air and water;
- design joints so that they have a high degree of resistance to the mechanisms of water entry - kinetic energy, surface tension, gravity, wind-assisted capillarity; and
- have an effective means of draining water from the system - usually via transoms rather than mullions.

Fundamentally therefore, with pressure equalised systems the outer face is sealed as tight as possible against rainwater, while the inner face is sealed as tight as possible against air inflow. Nearly all modern curtain walling systems are designed using the principles of pressure equalisation.

The performance of the air barrier is crucial in the success or failure of the pressure equalisation system. The barrier needs to be rigid in order to provide an air cavity of constant volume. Pulses of air pressure during gusting wind would take longer to dispel if the air volume changed, and if the seal were flexing under wind pressure it could also serve as a pump. (Rousseau, M.Z., *Facts and fictions of rainscreen walls*, article in *Construction Canada*, Vol. 32, no. 2, National Research Council, Canada, 1990)

The effects of wind pressure on buildings are complex, with corners and the highest parts of the building suffering sharp pressure gradients. On the side facing the wind, positive pressures are experienced, while high negative pressures are experienced on the leeward side. Such pressure variations can cause air currents behind the outer face, with air moving from areas of high pressure to areas of low pressure. These movements run contrary to the principles of pressure equalisation and so it is important that separate zones are created and sealed from one another. This maximises the chances of attaining pressure equalisation much faster.



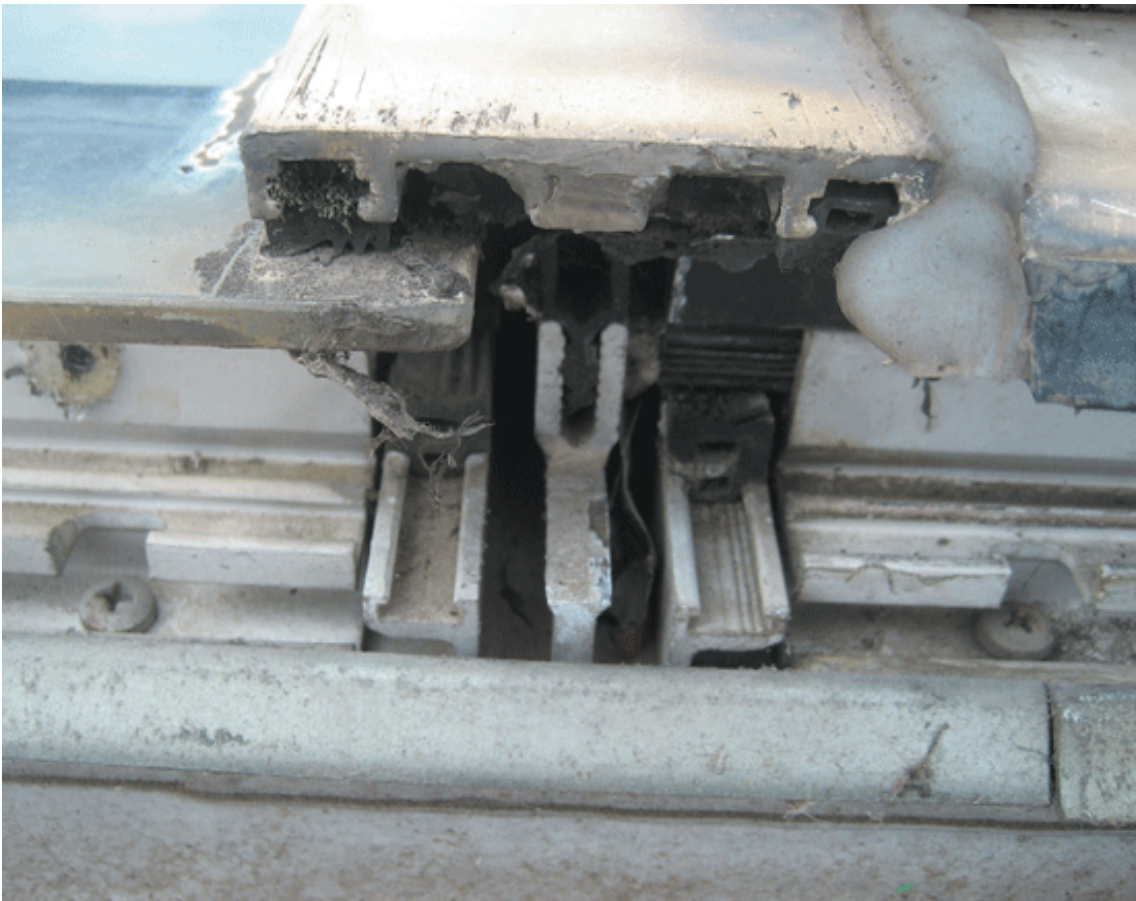
Method of pressure equalisation

Venting of the compartment permits the movement of air. Vents must be large enough to prevent water filming over them. The more effective the air barriers around the compartment, the smaller the area of venting. If the compartments are poorly sealed, larger amounts of air are needed, so more air must be drawn in from the outside.

Failure modes

When a component or system fails, it ceases to perform in accordance with the design intent. The failure mode is the way in which that failure occurs. Research suggests that water penetration is the most frequent failure mode, with buildability and durability being additional concerns (Layzell, J., *Failure mode and effects analysis in the cladding industry*, University of Bath, 1997).

The underlying factors that influence water penetration failures are generally due to a lack of knowledge of cladding installers, fabrication errors and contractual pressures such as late running on site.



Defective pressure plate system to a glazed rooflight. The clamping of the outer gaskets is insufficient and water penetrates the outer gasket into the drainage channel. Water in the drainage channel cannot escape owing to the lip of the roof upstand obstructing the channel at the bottom of the photograph. Note silicone top capping to try and alleviate leakage on the right of the photo.

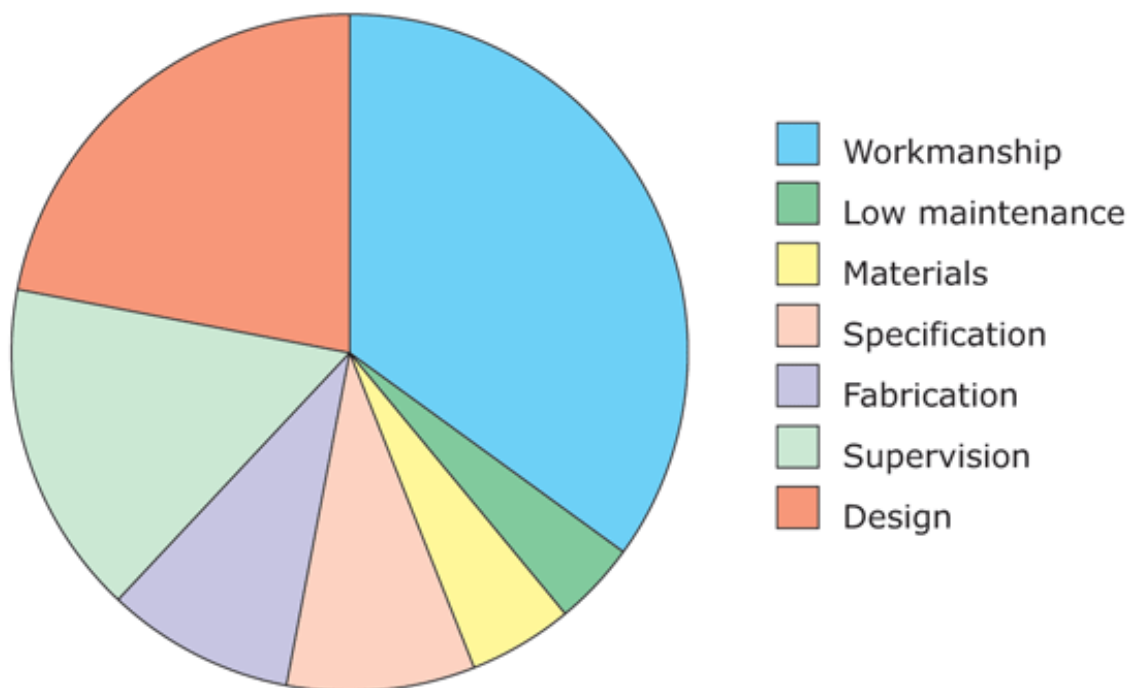
A study by Day in 1993 identified the different causes of failure; some 35 of his sample

reflected failings in workmanship, while 22% were failures in design (Day, R., *What effect will the publication of the new Standard and Guide to Good Practice have on CurtainWalling failures in the UK?*, 1993).

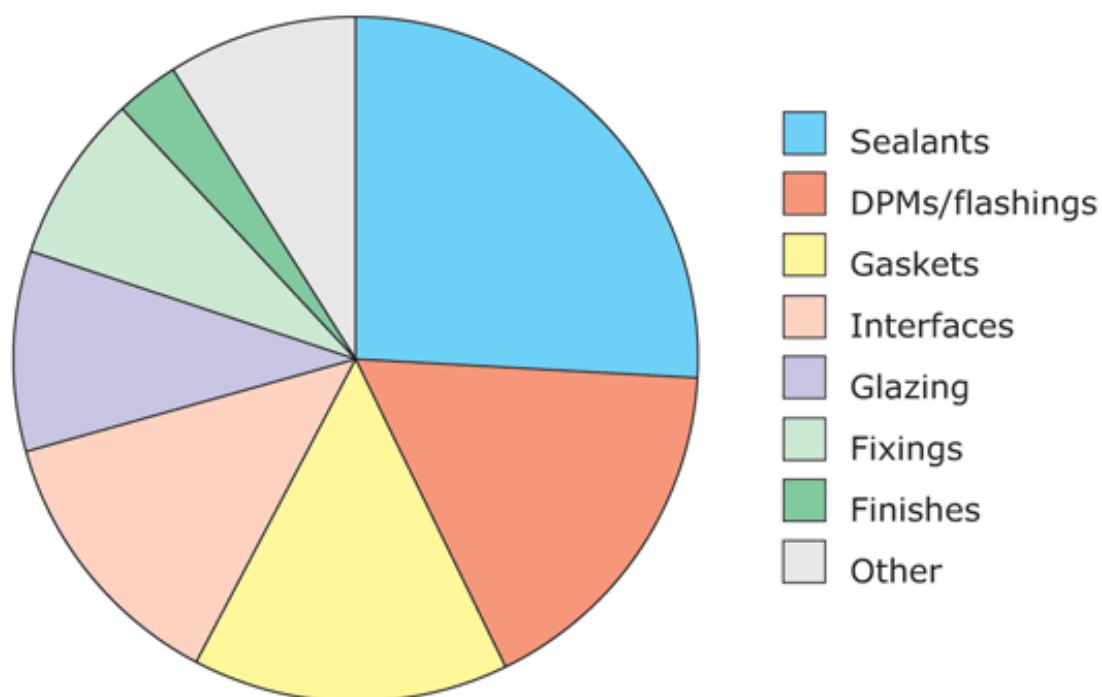
In 1997, a CWCT study (*Cladding Component Failures*) concluded that there were 5 main causes of watertightness failure :

- defective sealant joints;
- incorrect selection and sealing of corner joints in gaskets;
- mitre joints in windows;
- poorly sealed frame connections; and
- missing or sealed drainage holes and redundant fixing holes.

The same study identified sealants, gaskets and metal finishes as the components of cladding that were most likely to fail - often as a result of inappropriate or misguided selection and use.



Causes of curtain wall failures



Elements likely to cause problems

Identifying defects

The preceding pages have identified the main mechanisms of water ingress and the main systems of cladding. Knowledge of these principles can be put to use in diagnosing leakage problems in cladding systems that may be unfamiliar.

Analysis of leakage patterns is seldom straightforward, as water can travel some distance within framing members before it becomes manifest internally. A methodical approach is needed, together with the courage to open up.

Stage 1 - initial reconnaissance

According to Field Marshal Montgomery, 'time spent in reconnaissance is seldom wasted'. To identify the cause of cladding defects an initial inspection is essential. But the first inspection should not be a simple viewing of the problem, it should be tackled with some measure of science and methodology. For example, it would be little use inspecting a building only to agree with the occupant that the cladding is leaking - what is needed is an analysis of the locations and an understanding of the weather conditions that cause leaks, their duration and magnitude and whether there are any common patterns.

This may seem obvious, but it is surprising how many surveyors fail to establish patterns and locations of leaks - these can often help to pinpoint potential problems. For example, the initial

review may reveal that the leaks all occur at a certain level or always in line with a particular feature, a change in level or the interface between 2 different forms of cladding. Does orientation or level have a bearing? Are there any obvious defects? A quick check of the inner gaskets can reveal whether or not there is insufficient pressure - they should resist finger pressure, if they can be pushed inwards slightly, the external pressure plates may not be fixed to the correct torque.

Another common failing is to think in 2 dimensions rather than 3 - easier said than done, particularly as most graphical representations are drawn in 2 dimensions - usually because it is easier to do it that way.

Stage 2 - investigate the history and background information

Having gathered the initial information, attention needs to focus on identifying the type and design of the system. Here an examination of the operating and maintenance manuals should help. Unfortunately, in older buildings a lack of as-built information is common, but if it is possible to identify the manufacturer, it might be possible to obtain construction details, perhaps not of the actual building but at least of the system.

Many manufacturers have ceased trading, so if all else fails, archival research may yield further information. Journals such as the Architects Journal and Building often have building case files from back issues. These can provide some information as to the systems used, although this may be limited in terms of its reliability.

The purpose of this stage is to try and understand as much about the assembly process and design features as possible:

- How were joints to be formed?
- Was it mullion or transom drained?
- Are there rubber infiltration blocks?

and so on.

Stage 3 - site investigation

A visual inspection of the cladding installation will reveal useful information. However, for many systems it will be difficult to tell much without removal of at least the cover caps and probably some of the pressure plates too.

The object of this inspection is to check to see if the assembly of the frame, the fitting of setting blocks and the use of sealants is in accordance with the design and manufacturer's instructions. Check that pressure plates are installed the correct way up, i.e. that the slots for ventilation are not confused with the drainage slots. Check that the drainage paths are clear and not silted up and make sure that ad hoc amendments have not been made to drainage paths, such as the well intentioned but misinformed application of mastic sealant.

Water leakage can travel some distance such that the point of entry is not necessarily close to the point of egress. There are several methods of field testing that are generally recognised:

- Pressure spray in accordance with AAMA standard 501-94, *Methods of test for exterior walls*, American Architectural Manufacturers Association, USA.
- *Test methods for curtain walling*, CWCT, 1996.
- *Curtain walling. Watertightness site test*, BS EN 13051:2001.
- *Methods of testing windows. Watertightness test under static pressure*, BS

- 5368-2:1980.
- ASTM E 1105-96, *Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Curtain Walls and Doors by Uniform and Cyclic Static Air Pressure Difference.*

The CWCT and AAMA test regime is essentially similar.

Field tests, if executed methodically, can be a useful means of identifying leakage paths, but require care and patience. They also need an observer on the inside, preferably equipped with a 2-way radio for communication. If the performance of the cladding installation is in dispute, it is best to stay within the parameters set in the standards; if the matter is non-contentious then it is possible to relax some of the requirements as long as the basic methodology is followed.

However, the 2-pressure spray methods, AAMA 501-94 and CWCT are rigorous in their application; they are designed for testing site installations during construction and unless used with care and judgment will create leaks in joints and gaskets that would perform adequately in normal service conditions. Similarly, if leaks are produced during the test, you need to define whether or not the degree of leakage is acceptable (the result of extreme conditions) or whether remedial work is necessary.

Applying a pressure test to a feature that is not designed to deal with it (such as a rainscreen cladding) will inundate the system and drainage paths leading to a false result. It is usually possible to make something leak, but whether or not this produces the right answer depends on a careful and methodical approach. It is easy, for example, to form a pre-conceived view as to why a failure has occurred and then inadvertently make the evidence fit the theory.

The water spray test

Water is supplied via a compressor pump and break tank at a pressure of 220?20kPa to a suitable nozzle (CWCT specify a Monarch Type B-25, #6.030 nozzle), which produces a 30? spray cone. The cone is held perpendicular to the surface being tested at a distance of 300mm. A 5-minute test period is suggested for a length of component of 1.5m, i.e. passing the nozzle back and forth over 1.5m for a 5-minute period. This test uses around 110 litres of water over 5 minutes, so precautions may be needed if, for example, work is carried out above a pedestrian area or building entrance.

If no leakage is determined, the test can move on to the next 1.5m section and so on, starting at the bottom of the sample area or facade and gradually working up.

The process may identify a suspect area that can then be investigated, but if the exact position is unclear, masking tape can be applied over the joints and then progressively exposed to narrow down the area of defect.

If water leakage is determined it may be necessary to remove pressure plates and gaskets so that water paths can be examined.

The water spray test is not suitable for all types of cladding and for rainscreens or unsealed patent glazing systems. A spray bar test would be better as this reduces the pressure of water hitting the surface and produces a curtain of droplets flowing down the wall in a manner that replicates actual conditions. BS EN 13051 sets out the procedure to be followed.

The spray bar test

Using this system, a spray bar is fitted with suitable nozzles at a maximum of 400mm centres and suspended so that it is 250mm from the face of the cladding system. Water under mains pressure (2-3bar) is then applied and left to run for 30 minutes at a nozzle flow rate of 5 litres per minute. Unlike a water spray test, water should not be aimed directly at joints, but left to flow over them.

The cabinet test

This is usually restricted to windows and doors and involves constructing a sealed cabinet on one side of the component, extracting air from the cabinet to create negative pressure and then applying a water test (spray bar) to the outside. If the seals are defective, water can be drawn in as a result of the air pressure differential.

Cabinet tests are described in BS 5368-2. They are not particularly easy to carry out and rely upon the elimination of air leakage paths around the perimeter of the testing cabinet, or indeed around internal joints in the construction.

The cabinet is usually a timber frame sealed with a thick polythene liner, although an effective system can be constructed on site using timber battens to create a gap and a layer of polythene. The polythene has the advantage of allowing some measure of visibility, so that leakage paths can be tracked. The polythene must be held proud of the window or frame in question. Once assembled, air can be evacuated using a suction pump or industrial vacuum cleaner. A manometer is needed so that the pressure within the cabinet can be measured and monitored.

The test is usually carried out in 50Pa increments up to 200Pa, then 300Pa and finally in 150Pa increments up to the maximum pressure, which will need to be judged according to the specification of the glazing system.

See also Technical Update 2 - CWCT.

Checklist of critical points in rainscreen cladding

- Can water be discharged from the outer cladding face? Are there drips under external projections to throw water clear of the face?
- Is there a cavity behind the rainscreen of sufficient width to prevent capillary movements?
- Is there a continuous and effective air barrier? Are joints in the air barrier sealed and is it properly compressed?
- Can water drain away from the system? Is there a proper water path and is it clear from obstruction (moss, silt, etc.)?
- Is there adequate venting of the cavity? Does the area of venting take into account potential air leakages through the air barrier or changes of volume of the cavity (through flexible inner linings)?

- Is the facade compartmentalised to prevent lateral movement of air behind the rainscreen?
- Is the air barrier rigid or will it flex (and so vary the volume of the airspace)?
- Is the cladding sufficiently rigid to resist wind loads without excessive deflection?

The above checklist is based on *The rain screen wall system*, Ontario Association of Architect.

Condensation on metal framing

From a design point of view, the metal framing members of the curtain wall (or windows) should be maintained at a temperature slightly higher than the interior surface of the insulating glass unit (IGU), so that condensation will form on the glass first rather than its frame. Although temperatures differ according to the construction of the IGU, for a 6-12-6 unit the temperature of the frame will need to be greater than 10-12°C. (*Condensation on glazing frames*, Technical Update 3, CWCT)

The trend toward highly insulating units, using Low-e coatings, warm-edge technology and gas filled voids, tends to raise the inner surface temperature of the glass. According to CWCT, including a Low-e coating can raise the centre pane temperature to 15.2°C meaning that condensation could occur on the frame first unless a highly insulated frame design is chosen. Good practice dictates the use of better insulated framing for high efficiency insulating glass units.

Slight condensation in the form of misting towards the lower parts of the glass or window may be acceptable if it only occurs on a few occasions each year. However, if the condensation is severe enough to occur as running water, this is unsatisfactory and remedial measures should be contemplated. Such conditions may be the product of user abuse, (for example, removal of or failure to maintain ventilation) and/or high internal humidity due to processes or equipment failure - something that is difficult to control at the design stage.

Case study: Scottish office building

The building comprised a multi-tenanted office of circa 10,000m² arranged on a ground and 12 upper floors. The building's footprint reduced over 3 floors with a sloping glazed roof. The walls and roof were clad with a conventional drained and ventilated stick system which was designed to be mullion drained. Special shoes were built into the sloping roof mullions (rafters) to allow drainage into an aluminium gutter positioned at the transition of the sloping and vertical sections of cladding.

Water penetration was encountered at all floor levels to varying degrees, with most locations being confined to the areas below the sloping roof. Over the years, various unsuccessful investigations and trial repairs had led to accumulations of 'wet' mastic seals over pressure plates and gaskets. The window cleaning cradles had not been maintained and could not be used for inspection purposes; the only access being via a specially designed scaffold extending vertically, raking across the roof and then vertically upwards to the full height of the building.

Initially, selected cover strips and pressure plates were removed. Removal of the pressure plates released reservoirs of water that had been held in place by accumulations of silt and moss as well as the misguided applications of mastic, which in some cases had sealed drainage paths.

The aluminium gutter was tested and found to be watertight. However, the top lip of the gutter was formed of aluminium sheet held in place by the top pressure plate and gasket system to the vertical wall. When the pressure plate was properly compressed against the top glazing gasket, a good seal was formed between the glass and the gasket, but not between the gasket and the gutter. This led to water ingress into the system.

Tests using the AAMA procedure commenced at the bottom of the building, working upwards. Variable and random leakage patterns ensued suggesting that the basic concept of the system was sound, but that errors in construction might be responsible for localised faults.

To investigate fully, various IGUs were then removed and the glazing rebate inspected. According to the manufacturer's assembly details, a bead of sealant should have been applied to the mullion transom connections but this was missing in many areas. Also, the inner seals were assembled using pre-made corner pieces and straight gasket lengths. The contact faces between the corner pieces and the straight gaskets should have been sealed. They were not sealed, allowing water to be drawn into the building as a result of pressure differentials.

Attention was then turned to the drainage shoes at the base of the rafters. These shoes were designed to sit behind the face of the gutter and to drain into it via small pieces of hose cut in the side of the gutter. If the shoes leaked, water could only travel inside the building. This was solved by amending the gutter design and extending the rafters into the gutter itself to provide a fail-safe design.

Given the age of the glazing, and the need to remove all panels to achieve a satisfactory seal to the back gaskets and frame connections, all IGUs were replaced during the remedial works contract. The extent of water ingress was such that edge seals had been in contact with moisture for some period of time and so could have been compromised. All back gaskets were then sealed and fillet seals inserted at the transom and mullion connections. Following reassembly and reglazing, the entire areas were again field tested.

In summary, the water leakage was due to a combination of:

- poor initial assembly: a failure to seal the mullion and transom joints and a failure to make the back seals airtight;
- poor design in the detailing of the top gutter and the creation of a bespoke drainage shoe arrangement;
- poor maintenance: lack of the cradle meant that no cleaning had taken place, and the drainage paths had become compromised; and
- well-intentioned but inadequate repairs: failing to understand the mechanisms of water ingress and widespread use of sealant.

Case study: office building in coastal location

The building comprised a 1970s office block of rectangular plan constructed above an enclosed shopping centre near the coast. The cladding was steel framed, but single glazed. Coloured spandrel panels and fixed vision glass was held in place by external gaskets and pressure plates.

The cladding system leaked heavily. It relied upon a face seal with no provision for accidental leakage and no drainage paths. Over the years the system had fallen into disrepair and the frame had begun to corrode. The client needed to rectify the problem, but low rental income and poor cost recovery prospects had forced a series of compromises in repair strategies, usually involving copious amounts of sealant. Numerous pressure plates appeared to be loose and some cover caps were missing.

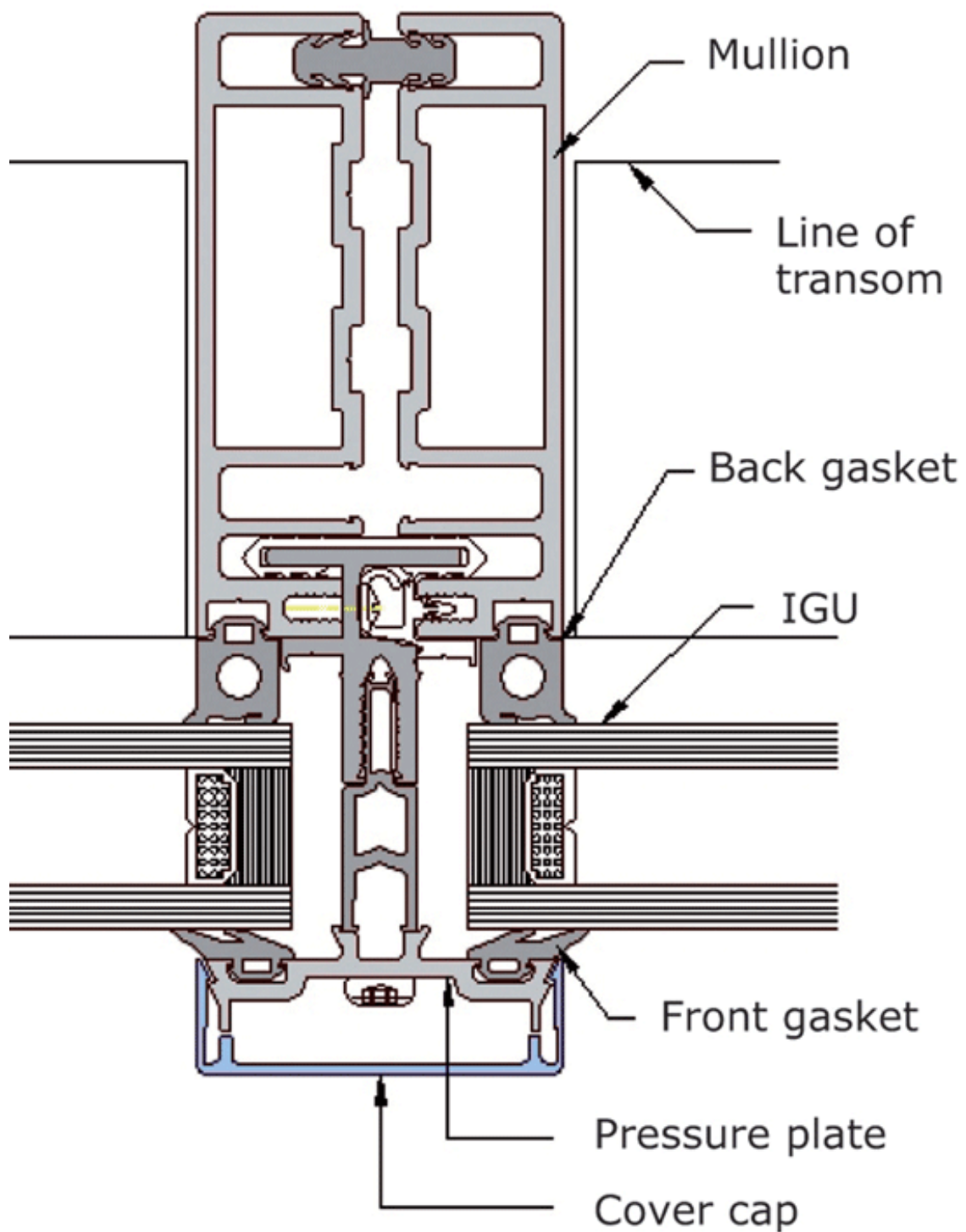
Water testing was unnecessary in this instance. Being single glazed, the provision of drainage channels around the glass was unnecessary for the purposes of durability, although if drainage could have been achieved it would have been beneficial.

In view of budgetary constraints, replacement of the cladding was impracticable and so a 2-stage external seal was provided. First, the old pressure plates were removed and discarded and the glass cleaned to remove all traces of the old mastic. Glass to metal joints were cleaned and sealed and a layer of butyl tape applied. New eagle gaskets and pressure plates could then be refitted, taking care to ensure that new fixing holes were drilled so that sufficient torque could be applied. The plates were then covered with drained cover caps. Internally, glass to metal joints around the inner seals were 'top capped' with silicone to form an internal air seal.

Problems with pressure plates

A typical aluminium pressure plate system is shown below. To exert effective pressure to the gasket, the pressure plate must be tightened to the correct torque (specified by the designer). Pressure plates are usually secured into the mullion screw fixing channel, but repeated insertion and renewal can lead to a weakening of the fixing and it is sometimes necessary to re-drill and re-secure a plate to achieve sufficient pressure.

In most cases a snap-on cover bead is provided. This is usually colour coated, although stainless steel varieties are available. The range of profiles is wide but they all rely upon a friction grip. The level of friction is governed by the closeness of fit (or interference) between the cover and the pressure plate. Both fittings are (usually) extrusions that will have been created using different die, the manufacture of which is subject to manufacturing tolerances. Problems can sometimes occur in batches supplied where the tolerance in dimension errs towards the low side, leading to a poor interference fit and therefore a loose pressure plate.



Section showing standard stick system with pressure plate

The figure below shows an aerofoil profile cover cap with a projection of around 100mm. The

manufacturer states that this type of cap must only be used in conjunction with a pressure plate of a particular profile. However, in this case, the aluminium profile has been omitted and instead a rectangular glass fibre pressure plate (resembling a strip of hardwood) has been used instead (an example of the installer mixing different components to save costs). The cover cap is a tight fit against the pressure plate, but there is little strength in the glass fibre to prevent the cap from being torn out.



Example of an incorrectly fitted cover plate - here the aerofoil-shaped cover has been wedged against a glass fibre pressure plate of inferior quality

The caps suffer from what might be loosely termed a 'Friday and Saturday Night' syndrome, as gangs of late night revellers have found that the covers at low level can be dislodged easily with a sharp kick. The more they are put back, the looser they become.

Of more concern is the fixity of the covers at higher level where lateral pressure (for example, from a window cradle) could pull the cover away. The pressure plates will need to be replaced with those of the correct specification.

Gaskets

Aside from silicone glazed systems, most cladding and window installations rely upon the proper performance of gaskets to maintain water and airtightness. To achieve a seal, the lip of the gasket must exert a pressure on the glass (lip pressure), usually circa 0.720N/mm to 1.760N/mm. (Nicastro, D., *Glazing system leaks*, article in *The Construction Specifier*, January 1996 issue, TheConstruction Specifications Institute, USA, 1996)

Gaskets are usually made from elastomeric materials such as butyl rubber which, although durable, degrade over time as a result of exposure to UV radiation, direct and reflected sunlight, and ozone. Volatile compounds in the gasket are driven off (out-gassing) by these mechanisms with the result that they eventually suffer shrinkage in volume and or brittleness. Often gasket materials are stretched as they are installed, with the result that they gradually shrink back into shape, opening corner or butt joints in the process.

It is better to anticipate some water leakage through the outer gasket system and to take steps to drain it away harmlessly to the outside. To an extent therefore, loss of lip pressure can be partly accommodated. However, if the inner or air seal gasket is also defective, there is a risk that water leakage can take place.

Replacement of the outer gaskets may be possible, but replacement of the air seals will probably involve deglazing (unless they are harder wedge gaskets - used when the glass in windows is bead glazed). An alternative to gasket replacement in these circumstances is the provision of a wet seal or top cap over the lip of the gasket.

Given accurate and careful workmanship a wet seal can be a useful expedient, but it should only be considered if the glazing system and its weaknesses have been properly understood. Unfortunately, the application of mastic is often seen as a quick fix, carried out by inexperienced applicators. For success, Nicastro recommends:

- Use only silicone sealant that can be shown to be compatible with the original gasket.
- Use black sealant and ensure that it is applied by trained applicators.
- Prior to application, clean all contact surfaces with solvent cleaner.
- Maintain a minimum cross section of fillet of 6mm and a contact area on each face of 6mm.

Wet sealing can make a marginal difference to the shading of the perimeter of the glass and carries a slight, possibly theoretical, risk of increasing thermal stress between the centre of a pane and its perimeter.

Sealant failure and reversion

Sealants are an essential part of modern construction and have a crucial role to play in the design of weathertight joints. ASTM defines a sealant in building terms as 'a material which has the adhesive and cohesive properties needed to form a seal'. Joints can fail as a result of poor design, workmanship or the incorrect choice of sealant for the job.

Broadly, there are several basic types of sealant failure:

- adhesion failure;
- cohesion failure;
- delamination failure; and
- reversion.

Adhesion failure is a loss of bond between the sealant and the surface to which it is applied. The failure mode is characterised by a visible separation along the line of the interface - this is termed interfacial adhesion failure. Most sealants require a primer or at least clean surfaces free of contamination by water, frost oil or dust. Omitting the primer and or failing to clean the contact surfaces prior to application may cause adhesion failure.

Cohesion failure is characterised by tears or splits within the body of the sealant, while the bonding faces remain intact. In these cases that the sealant has probably been overstressed either as a result of poor joint design or a loss of flexibility. Joint profile is important - it is generally accepted that for a sealed joint to work it should be bonded on 2 faces only. However, the thickness of the joint is also relevant. A thinner joint will possess more flexibility than a thicker one, but equally will be more vulnerable to the effects of exposure to UV light and the ensuing degradation.

The profile of the sealant joint influences the performance of the sealant. For example, the triangular fillet profiles that are often found around windows perform badly in comparison to rectangular joints, particularly if they are concave. (*Defects in buildings*, HMSO, London, 1989)

Sealants are either designated by categories of $\pm 12.5\%$, $\pm 50\%$ or $+100\%/-50\%$ of initial movement capability. Joint movement on a percentage basis is also affected by the size of the joint. For example, a 6mm movement on a 13mm wide joint will be twice as much on a percentage basis as a 6mm movement on a 25mm wide joint. Therefore, narrower joints are more prone to cohesive failure of the sealant. Many polyurethane sealants will stiffen and show a loss in movement capability over time, making them vulnerable to cohesion failure.

Delamination failure occurs when the sealant overstresses a coating or material to which it is applied. For example, the bond between a mastic seal and a paint film may be perfect, but joint movement produces a stress that is too great for the paint/substrate bond to accommodate. Polyurethane sealants also have the inherent property of increasing 2- to 3-fold in sealant modulus in cold temperatures, making them stiffer and less flexible, thereby increasing joint stress. (*EIFS Restoration Guide*, Dow Corning Corporation, USA, 2002)

Similar effects can occur when the substrate becomes softened as a result of trapped moisture. Polyurethane backing rods can absorb moisture and hold it in contact with certain vulnerable materials, such as acrylic coatings applied to external rendering systems. Therefore, external acrylic coatings should be stopped short of a joint before the application of sealant. Such failures are termed interphasal failures.

Reversion failures and deterioration

Reversion is a term used to describe deterioration in the elastomeric properties of a sealant. (Some authorities term this as a loss of rheological properties.) Reversion is a difficult subject as there are mixed scientific opinions as to its cause. Essentially, reversion is the conversion of a sealant from a cured to an uncured state, usually initiated by high temperatures, UV exposure and water vapour. Experimentation has revealed conflicting evidence as to the process. The colour of the sealant may also play a part as a result of the energy absorbing capabilities of the different pigments.

Sealant reversion can be tested fairly easily by taking a sample of the material and rolling it into a ball. If the sealant holds its new shape, it has lost its rheological properties and can be termed reverted.

Another form of reversion, again easily identified, is the reversion of silicone sealant back to a powder form. Typically, this can be seen to affect the thin joints in stone cladding and is probably the result of high temperature and moisture.

Sealants such as multi-component polyurethanes are organic materials that degrade in UV light. The degradation results in loss of flexibility and movement capability, ultimately ending in reversion.

Hardening of the sealants over time can cause a greater stress to be exerted at the bonding face with each extension and compression cycle. If the stress exceeds the allowable stress in the background material, localised adhesion failure can occur.

Cladding and building movements

Cladding systems must be able to tolerate expansion and contraction in the glass and framing members together with normal movements, deflections and long-term axial shrinkages in the structure to which it is fixed. The effects of wind sway, differential settlements and elastic deformations also need to be factored into the design of both framing and fixing systems.

Thermal movements in frames must be accommodated. If there is no provision for movement in the frame design, stresses will build up in the profile. The less the provision for movement, the greater the stress and the greater the propensity for deflection or buckling. For northern climes, it is usual to apply a rule of thumb of 1mm movement provision for every 1m of length of component and to calculate stress with this in mind. The typical range of UK temperatures for design purposes could be -25°C to +80°C. If the stress in the element in question is allowed to exceed its permissible stress, failure can occur.

Curtain wall systems are usually fixed by way of brackets back to the floor slab. Deflections at the edge of the floor slab are usually calculated at design stage - possibly span $\times$ 360 or span $\times$ 500, giving 18-25mm movement for a span of 9,000mm. The system must accommodate this movement, but to achieve this it may be necessary to incorporate split transoms as well as sliding mullion joints. A split transom usually incorporates twin overlapping pressings and an internal concertina gasket.

Concrete structures suffer from long term creep - axial shrinkage such that movement of as much as 0.1% can be expected, for example, 30mm for every 30m in height (*Wall cladding defects and their diagnosis*, BRE Digest 217, 1978). Movement will become progressively greater towards the top of the building and must be accommodated in the cladding and glazing systems to prevent stress formation.

Spandrel panels

A typical fully glazed curtain wall system will comprise a mixture of vision glass and opaque 'look-a-like' spandrel panels used to conceal the floor edge construction, ceiling void, heating plant, etc.

Spandrel panels pose specific difficulties since the dark colours used create high temperatures in the glass or the void behind it: 90°C is not unknown. Broadly, there are 2 methods of creating look-a-like effects:

- adding an opacifier to the rear face of the glass; and
- creating a dark painted void behind the glass - a shadow box.

Opacifiers are usually added to the fourth face of an IGU, being either a polyester or (mylar) film or alternatively a silicone spray treatment. Ceramic frits are also used. High temperatures can cause applied opacifier films to blister, while oil or latex paints are short lived as UV rays break down the molecular bond between the paint and the glass. Most opacifier failures are associated with shrinkage, weathering, oxidation or adhesion failures in films that are sensitive to moisture, heat or light.

Annealed glass can be used for spandrel panels, but only if certain criteria are met. A ventilated void would be essential to remove warm air, and glass edges could also be polished to reduce the risk of thermal stress fractures.

If annealed glass were to fail, sharp shards of glass could fall out and so some form of scrim would need to be applied to the back face. Given the uncertainties of performance, heat strengthened or toughened glass is a more likely choice for spandrel panels. Heat strengthened glass can deal with higher working temperatures than annealed glass, although it can still shatter into shards. However, it does display less distortion due to roller wave.

Shadow box construction is difficult to achieve in practice and creates further problems of condensation control and heat build-up. High temperatures in the void behind the glass can permit volatile compounds to gas off and cause staining to the back of the glass when they condense. (*Spandrel Panel Glazing*, ATS-124, Pilkington Glass, 2005.)

Cleaning equipment

A detailed study of the various types of window cleaning installations is beyond the scope of this section. However, there are matters that need to be considered in conjunction with cladding design.

For buildings over 30m in height, a fixed installation is generally essential, but cradles will need additional attachment points such as slotted mullions, button guides, tie bolts or pins.

Typical maintenance standards for curtain walling demand cleaning of the metalwork up to 5 times a year. However, most installations are suitable in conditions where windspeed does not exceed 12m/sec. While this condition may be fine at ground level, conditions may be very different at upper stories in a tall building. If it takes 5 to 10 days to access all parts it is unlikely that weather conditions will be favourable on this number of days on 5 occasions throughout the year, so maintenance and inspection provisions may be compromised.



Roller wave distortions in toughened glass



Cladding systems must be cleaned regularly to prevent deterioration. Specialist cleaning may help to improve the finish on this colour coating but it is probably too far gone to be restored to its original condition.